Homework Assignment 2

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Census Data

1) Use the tidycensus package to download American Community Survey five-year estimates from 2019-2023 in California at the County level for the following variable, including spatial data: a.) Household Median Income = "B19013_001"

```
## Getting data from the 2019-2023 5-year ACS

## Downloading feature geometry from the Census website.  To cache shapefiles for use in future sessions

##   |                                                                              |
```

```
##                           geometry
## 1  MULTIPOLYGON (((-122.0172 3...
## 2  MULTIPOLYGON (((-122.0023 3...
## 3  MULTIPOLYGON (((-118.2028 3...
## 4  MULTIPOLYGON (((-118.2196 3...
## 5  MULTIPOLYGON (((-118.4388 3...
## 6  MULTIPOLYGON (((-118.2202 3...
## 7  MULTIPOLYGON (((-118.2392 3...
## 8  MULTIPOLYGON (((-118.2379 3...
## 9  MULTIPOLYGON (((-118.2287 3...
## 10 MULTIPOLYGON (((-118.5023 3...

## Getting data from the 2019-2023 5-year ACS
```

2) Take a glimpse at your data.

```
#Glimpse at data set w/ "NAME" column separated & joined files

glimpse(ca.tracts)
```

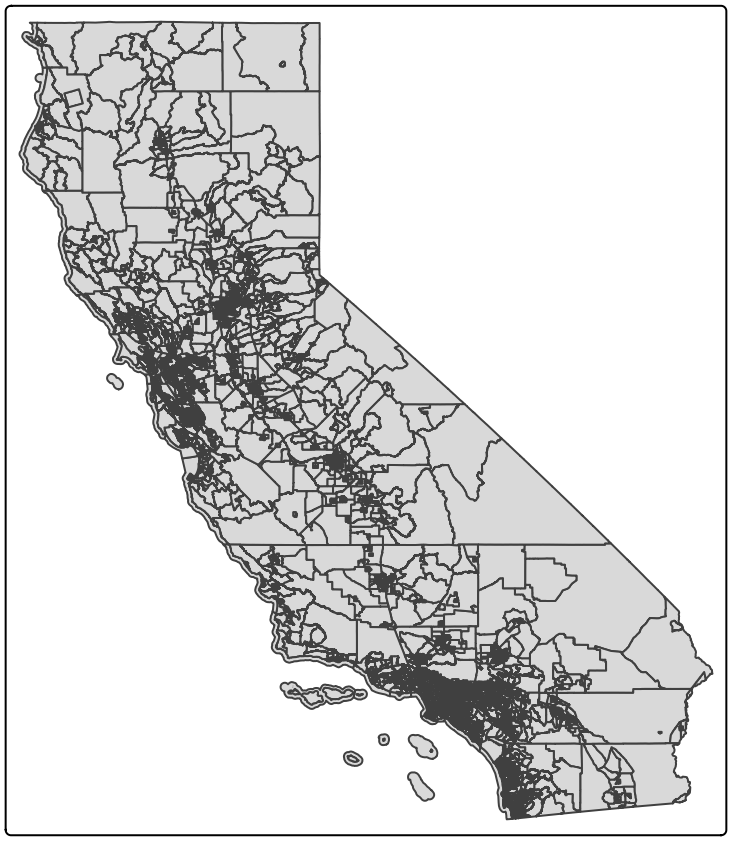
```
## Rows: 9,129
## Columns: 18
## $ GEOID    <chr> "06001442700", "06001442800", "06037204920", "06037205110", "~
## $ Tract    <chr> "Census Tract 4427", "Census Tract 4428", "Census Tract 2049.~
## $ County   <chr> "Alameda County", "Alameda County", "Los Angeles County", "Lo~
## $ State    <chr> "California", "California", "California", "California", "Cali~
## $ tpoprE   <dbl> 2998, 3087, 2459, 3690, 3402, 3409, 3490, 8179, 2290, 5998, 4~
## $ tpoprM   <dbl> 475, 386, 387, 446, 438, 624, 605, 350, 509, 1024, 578, 621, ~
## $ nhwhiteE <dbl> 921, 779, 68, 14, 236, 15, 500, 1379, 138, 2685, 2306, 11, 81~
## $ nhwhiteM <dbl> 168, 183, 33, 14, 131, 24, 264, 165, 49, 831, 535, 27, 73, 18~
## $ nhblkE   <dbl> 119, 12, 49, 8, 65, 8, 17, 2633, 18, 18, 93, 26, 2, 352, 1086~
## $ nhblkM   <dbl> 115, 49, 51, 12, 93, 10, 75, 344, 22, 19, 97, 41, 4, 264, 679~
## $ nhasnE   <dbl> 1577, 1671, 0, 0, 123, 70, 1030, 238, 151, 920, 665, 13, 29, ~
## $ nhasnM   <dbl> 469, 356, 14, 14, 164, 104, 235, 76, 82, 563, 176, 21, 45, 26~
## $ hispE    <dbl> 308, 561, 2331, 3642, 2978, 3303, 1919, 3541, 1932, 2322, 134~
## $ hispM    <dbl> 81, 181, 387, 446, 481, 628, 519, 285, 519, 433, 265, 618, 18~
## $ NAME     <chr> "Census Tract 4427; Alameda County; California", "Census Trac~
## $ medincE  <dbl> 199154, 180800, 70500, 52262, 110967, 28516, 60703, 148661, 4~
## $ medincM  <dbl> 30525, 28293, 15698, 5659, 22149, 3246, 28578, 39223, 19295, ~
## $ geometry <MULTIPOLYGON [°]> MULTIPOLYGON (((-122.0172 3..., MULTIPOLYGON (((~
```

3) Make a map of median income by County.

```
tmap_mode("plot") #Best tmap mode for knitting file
```

```
## i tmap modes "plot" - "view"
## i toggle with ‘tmap::ttm()‘
```

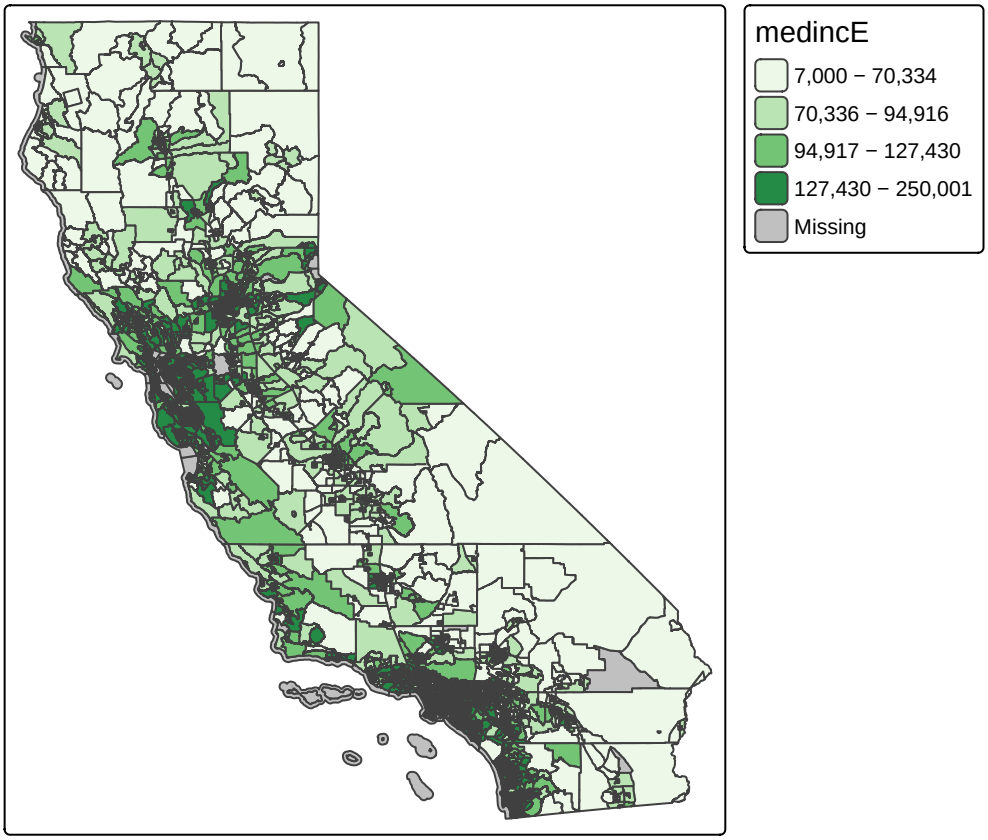
```
tract_map <- tm_shape(ca.tracts) +   tm_polygons()

tract_map
```



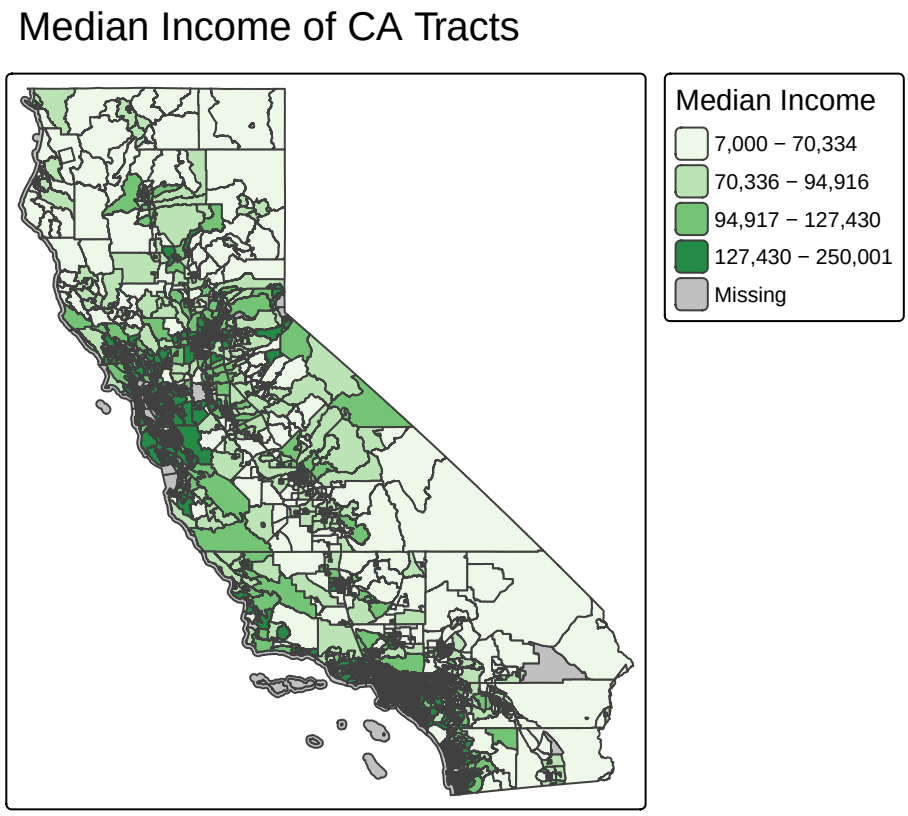
4) Now can you change the color palette?

```
#Changed the tracts to be a green color scheme
tm_shape(ca.tracts) +
  tm_polygons(fill = "medincE",
  fill.scale = tm_scale_intervals(style = "quantile", n=4, values = "Greens"))
```

```
## [cols4all] color palettes: use palettes from the R package cols4all. Run
## `cols4all::c4a_gui()` to explore them. The old palette name "Greens" is named
## "brewer.greens"
## Multiple palettes called "greens" found: "brewer.greens", "matplotlib.greens". The first one, "brewe
```


medincE
7,000 − 70,334
70,336 − 94,916
94,917 − 127,430
127,430 − 250,001
Missing

5) Can you add a nice title and add a legend?



Raster Data

6) Import land surface temperature data with the terra package.

```
## class       : SpatRaster
## size        : 1056, 1153, 1  (nrow, ncol, nlyr)
## resolution  : 0.008983153, 0.008983153  (x, y)
## extent      : -124.4885, -114.131, 32.528, 42.01421  (xmin, xmax, ymin, ymax)
## coord. ref. : lon/lat WGS 84 (EPSG:4326)
## source      : LST_med_ca.tif
## name        : LST_Day_1km_median

## Coordinate Reference System:
##   User input: WGS 84
##   wkt:
## GEOGCRS["WGS 84",
##     ENSEMBLE["World Geodetic System 1984 ensemble",
##         MEMBER["World Geodetic System 1984 (Transit)"],
##         MEMBER["World Geodetic System 1984 (G730)"],
##         MEMBER["World Geodetic System 1984 (G873)"],
##         MEMBER["World Geodetic System 1984 (G1150)"],
##         MEMBER["World Geodetic System 1984 (G1674)"],
```

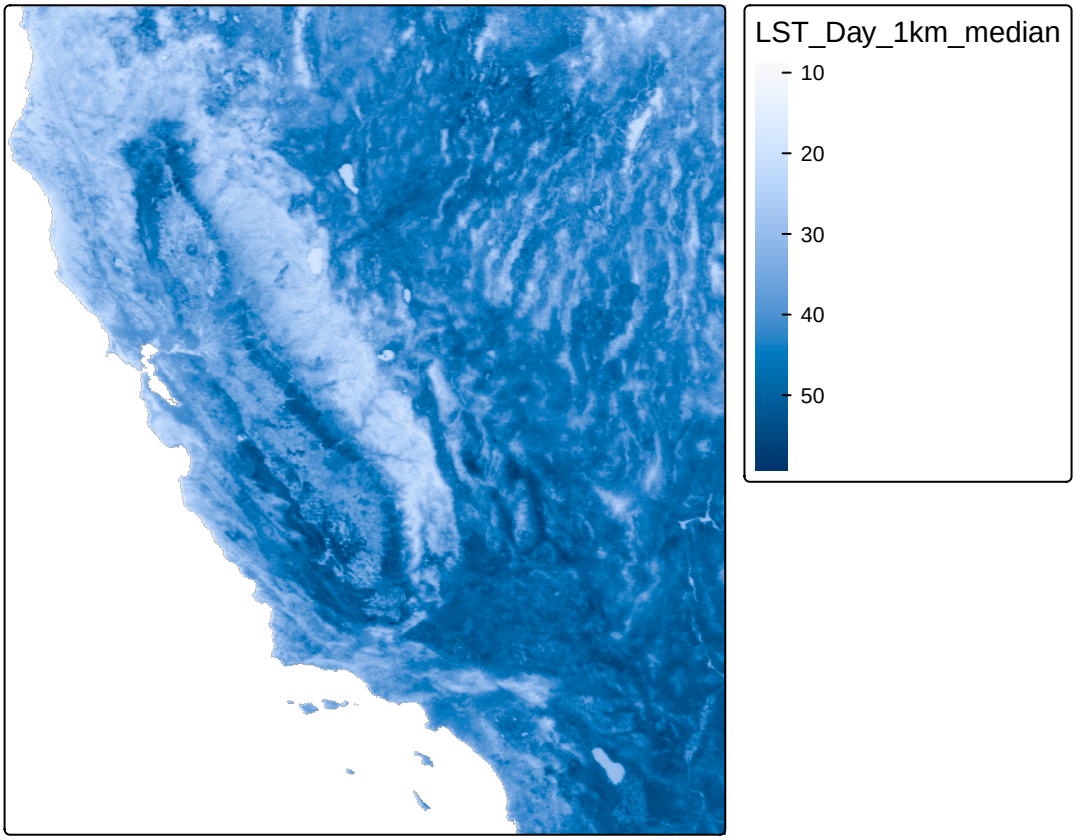
```
##         MEMBER["World Geodetic System 1984 (G1762)"],
##         MEMBER["World Geodetic System 1984 (G2139)"],
##         MEMBER["World Geodetic System 1984 (G2296)"],
##         ELLIPSOID["WGS 84",6378137,298.257223563,
##             LENGTHUNIT["metre",1]],
##         ENSEMBLEACCURACY[2.0]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
##         AREA["World."],
##         BBOX[-90,-180,90,180]],
##     ID["EPSG",4326]]
```

7) Make a map of land surface temperature across California

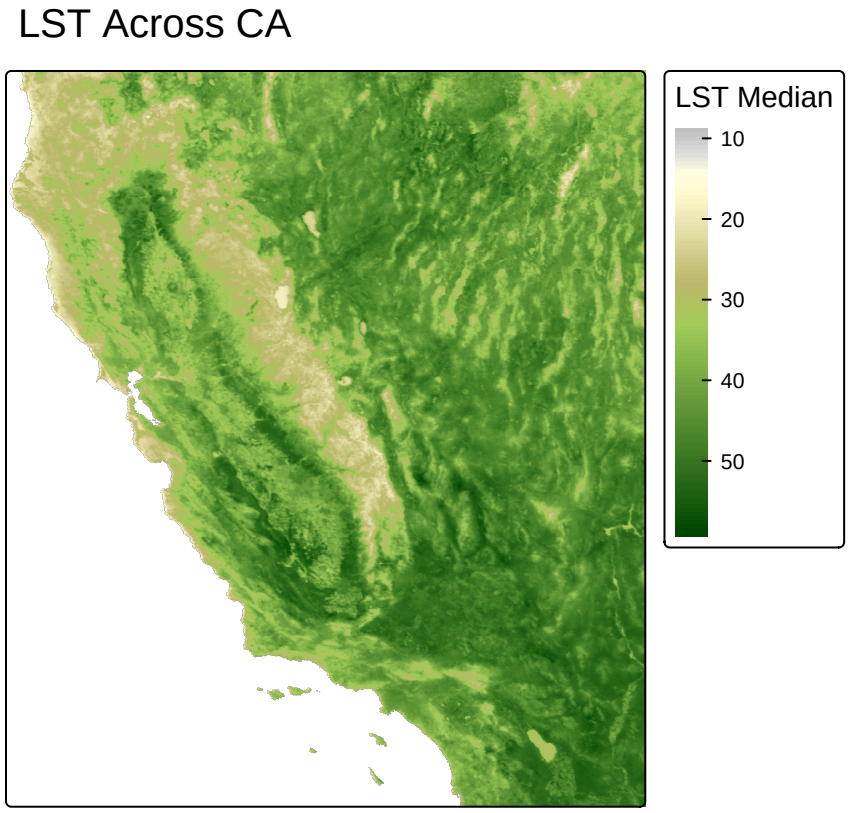
```
tmap_mode("plot")
```

```
## i tmap modes "plot" - "view"
```

```
LST_map <-  tm_shape(LST_raster) +
  tm_raster(col.scale = tm_scale_continuous(),
            col.legend = tm_legend(outside = TRUE))
LST_map
```




```
# palette for plotting
palette_ndvi <- c("#BFBFBF","#DBDBDB","#FFFFE0","#FFFACC","#EDE8B5","#DED99C","#CCC782","#BDB86B","#B0C

LST_map <-  tm_shape(LST_raster) +
  tm_raster(col.scale = tm_scale_continuous(values = palette_ndvi),
            col.legend = tm_legend(title = "LST Median", outside = TRUE)) + tm_title(text = "LST Across
           size = 1.25,
           position = tm_pos_out("center","top"))
LST_map
```

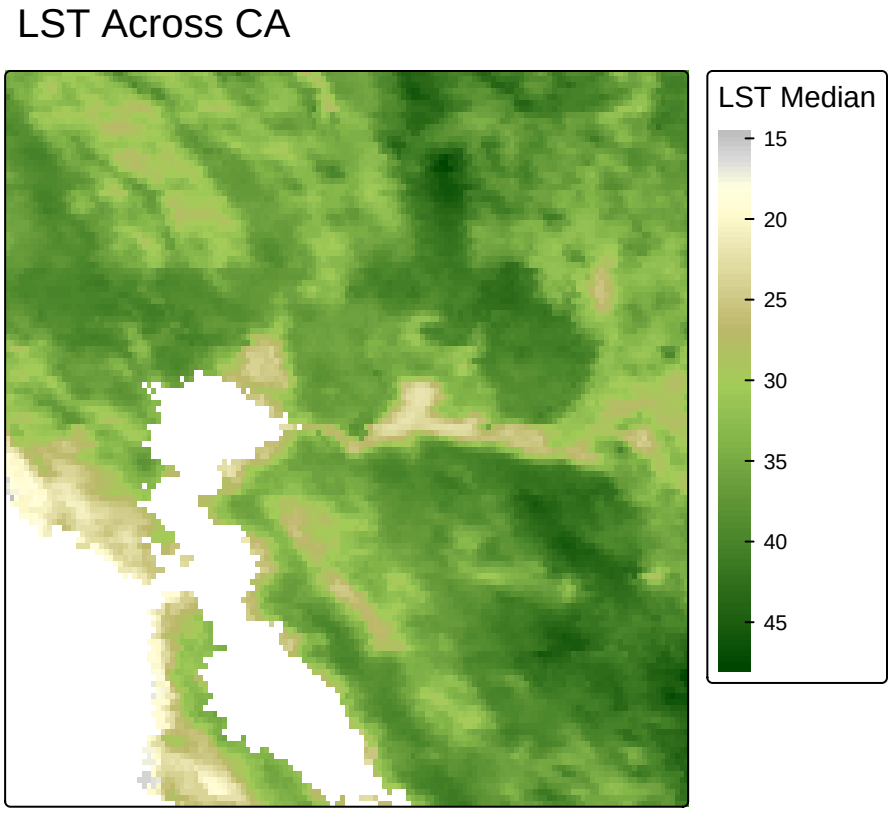


8) Crop to these coordinates: ext(-122.8, -121.5, 37.5, 38.6) and make a map again.

```
#Crop coordinates
LST_crop <-crop(LST_raster, ext(-122.8, -121.5, 37.5, 38.6))

#Plot LST Raster Map w/ Cropped Coordinates
tmap_mode("plot")
```

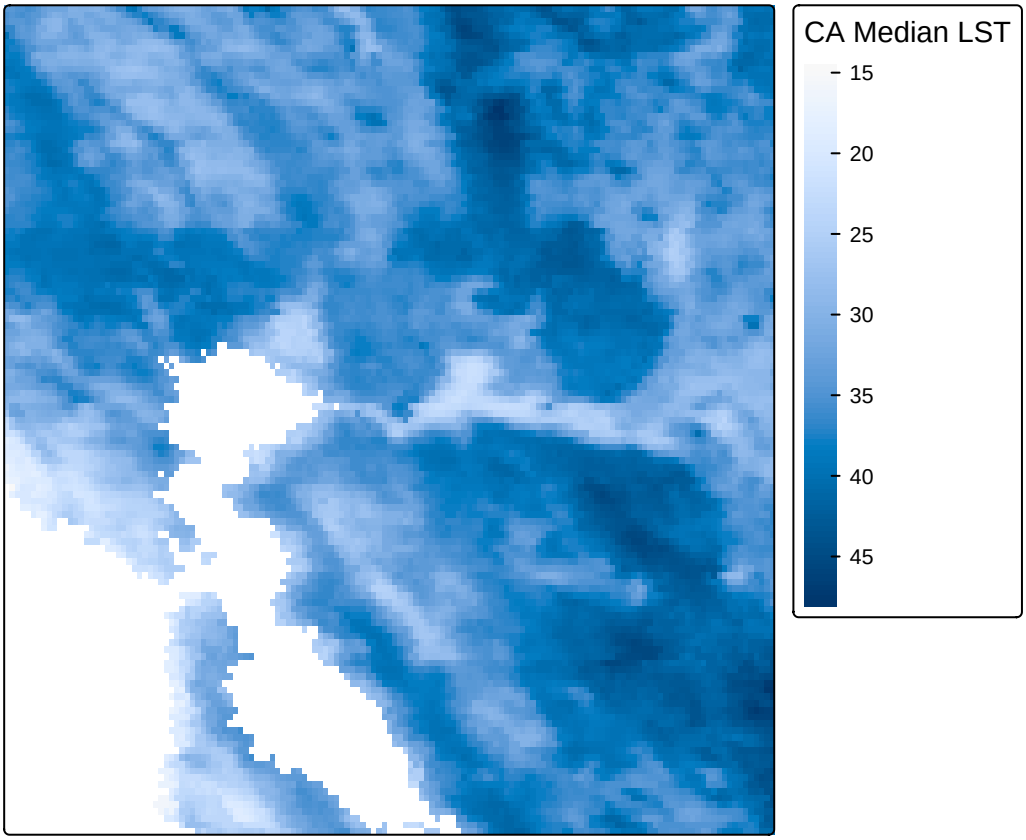
```
## i tmap modes "plot" - "view"
```

```
LST_ca <- tm_shape(LST_crop) + tm_raster(col.scale = tm_scale_continuous(values = palette_ndvi), col.leg

LST_ca
```



8) Continued...

```
#Use negative values to distinguish between land and water

lst_rast_neg <- app(LST_crop, fun=function(x){ x[x<= 0] <- -1; return(x)})

lst_crs_map_neg <- tm_shape(lst_rast_neg) +
  tm_raster(col.scale = tm_scale_continuous(),
            col.legend = tm_legend(title = "CA Median LST", outside = TRUE))

lst_crs_map_neg
```



Overlay Cancer and Land surface temperature Data

9) Bring in our old CAdata ovarian cancer data set

```
download.file("https://raw.githubusercontent.com/pjames-ucdavis/SPH215/refs/heads/main/CAdata.rds", "CA

CAdata <- readRDS("CAdata.rds")

ca_pts <- CAdata
ca_proj <- "+proj=lcc +lat_1=40 +lat_2=41.66666666666666
            +lat_0=39.33333333333334 +lon_0=-122 +x_0=2000000
            +y_0=500000.0000000002 +ellps=GRS80
            +datum=NAD83 +units=m +no_defs"

ca_pts <- st_as_sf(CAdata, coords=c("X","Y"), crs=ca_proj)
```

10) Overlay a map of CAdata with the geocoded participant addresses in red on top of a map of the land surface temperature data for the whole state (not just what you cropped to).

```
#Compare the data sets CRS
st_crs(ca_pts)
```

```
## Coordinate Reference System:
##   User input: +proj=lcc +lat_1=40 +lat_2=41.66666666666666
##               +lat_0=39.33333333333334 +lon_0=-122 +x_0=2000000
##               +y_0=500000.0000000002 +ellps=GRS80
##               +datum=NAD83 +units=m +no_defs
##   wkt:
## PROJCRS["unknown",
##     BASEGEOGCRS["unknown",
##         DATUM["North American Datum 1983",
##             ELLIPSOID["GRS 1980",6378137,298.257222101,
##                 LENGTHUNIT["metre",1]],
##             ID["EPSG",6269]],
##         PRIMEM["Greenwich",0,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8901]]],
##     CONVERSION["unknown",
##         METHOD["Lambert Conic Conformal (2SP)",
##             ID["EPSG",9802]],
##         PARAMETER["Latitude of false origin",39.3333333333333,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8821]],
##         PARAMETER["Longitude of false origin",-122,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8822]],
##         PARAMETER["Latitude of 1st standard parallel",40,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8823]],
##         PARAMETER["Latitude of 2nd standard parallel",41.6666666666667,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8824]],
##         PARAMETER["Easting at false origin",2000000,
##             LENGTHUNIT["metre",1],
##             ID["EPSG",8826]],
##         PARAMETER["Northing at false origin",500000,
##             LENGTHUNIT["metre",1],
##             ID["EPSG",8827]]],
##     CS[Cartesian,2],
##         AXIS["(E)",east,
##             ORDER[1],
##             LENGTHUNIT["metre",1,
##                 ID["EPSG",9001]]],
##         AXIS["(N)",north,
##             ORDER[2],
##             LENGTHUNIT["metre",1,
##                 ID["EPSG",9001]]]]
```

```r
st_crs(lst_rast_neg)
```

```
## Coordinate Reference System:
##   User input: WGS 84
##   wkt:
## GEOGCRS["WGS 84",
##     ENSEMBLE["World Geodetic System 1984 ensemble",
##         MEMBER["World Geodetic System 1984 (Transit)"],
##         MEMBER["World Geodetic System 1984 (G730)"],
##         MEMBER["World Geodetic System 1984 (G873)"],
##         MEMBER["World Geodetic System 1984 (G1150)"],
##         MEMBER["World Geodetic System 1984 (G1674)"],
##         MEMBER["World Geodetic System 1984 (G1762)"],
##         MEMBER["World Geodetic System 1984 (G2139)"],
##         MEMBER["World Geodetic System 1984 (G2296)"],
##         ELLIPSOID["WGS 84",6378137,298.257223563,
##             LENGTHUNIT["metre",1]],
##         ENSEMBLEACCURACY[2.0]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
##         AREA["World."],
##         BBOX[-90,-180,90,180]],
##     ID["EPSG",4326]]
```

```r
#Transform - Make them on the same projection
ca_pts_proj <- st_transform(ca_pts,st_crs(lst_rast_neg))

#Compare once more
st_crs(ca_pts_proj)
```

```
## Coordinate Reference System:
##   User input: WGS 84
##   wkt:
## GEOGCRS["WGS 84",
##     ENSEMBLE["World Geodetic System 1984 ensemble",
##         MEMBER["World Geodetic System 1984 (Transit)"],
##         MEMBER["World Geodetic System 1984 (G730)"],
##         MEMBER["World Geodetic System 1984 (G873)"],
##         MEMBER["World Geodetic System 1984 (G1150)"],
##         MEMBER["World Geodetic System 1984 (G1674)"],
##         MEMBER["World Geodetic System 1984 (G1762)"],
##         MEMBER["World Geodetic System 1984 (G2139)"],
##         MEMBER["World Geodetic System 1984 (G2296)"],
```

```
##         ELLIPSOID["WGS 84",6378137,298.257223563,
##             LENGTHUNIT["metre",1]],
##         ENSEMBLEACCURACY[2.0]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
##         AREA["World."],
##         BBOX[-90,-180,90,180]],
##     ID["EPSG",4326]]
```

```
st_crs(lst_rast_neg)
```

```
## Coordinate Reference System:
##   User input: WGS 84
##   wkt:
## GEOGCRS["WGS 84",
##     ENSEMBLE["World Geodetic System 1984 ensemble",
##         MEMBER["World Geodetic System 1984 (Transit)"],
##         MEMBER["World Geodetic System 1984 (G730)"],
##         MEMBER["World Geodetic System 1984 (G873)"],
##         MEMBER["World Geodetic System 1984 (G1150)"],
##         MEMBER["World Geodetic System 1984 (G1674)"],
##         MEMBER["World Geodetic System 1984 (G1762)"],
##         MEMBER["World Geodetic System 1984 (G2139)"],
##         MEMBER["World Geodetic System 1984 (G2296)"],
##         ELLIPSOID["WGS 84",6378137,298.257223563,
##             LENGTHUNIT["metre",1]],
##         ENSEMBLEACCURACY[2.0]],
##     PRIMEM["Greenwich",0,
##         ANGLEUNIT["degree",0.0174532925199433]],
##     CS[ellipsoidal,2],
##         AXIS["geodetic latitude (Lat)",north,
##             ORDER[1],
##             ANGLEUNIT["degree",0.0174532925199433]],
##         AXIS["geodetic longitude (Lon)",east,
##             ORDER[2],
##             ANGLEUNIT["degree",0.0174532925199433]],
##     USAGE[
##         SCOPE["Horizontal component of 3D system."],
##         AREA["World."],
##         BBOX[-90,-180,90,180]],
##     ID["EPSG",4326]]
```

```
#Overlay Maps - w/o cropped coordinates
tmap_mode("plot")
```

```
## i tmap modes "plot" - "view"
```

```
lst_cancer_map <-  tm_shape(LST_raster) +
  tm_raster(col.scale = tm_scale_continuous(),
            col.legend = tm_legend(title = "CA Median LST", outside = TRUE)) +
  tm_shape(ca_pts_proj) +
  tm_dots(fill = "red", fill_alpha = 0.5, size = 0.3)

lst_cancer_map
```